

Agile Handbook – Batch 3

Kanban, XP, and Lean Software
Development

Kanban Overview

- Visual workflow management method
- Uses a Kanban board with columns (To Do, In Progress, Done)
- Focuses on continuous delivery and flow efficiency
- Helps teams visualize work and limit work in progress

Kanban Principles

- Visualize the workflow
- Limit Work in Progress (WIP)
- Manage flow and measure performance
- Make process policies explicit
- Improve collaboratively and evolve experimentally

Kanban vs Scrum

Aspect	Kanban	Scrum
Cadence	Continuous flow	Time-boxed Sprints
Roles	No required roles	Scrum Master, PO, Team
Changes	Changes anytime	Changes only between Sprints
WIP Limits	Core principle	Not enforced
Metrics	Lead time, cycle time	Velocity, burndown

Kanban Real-world Use Cases

- IT Operations and Support Teams
- Marketing and Content Production
- HR Recruitment Pipelines
- Maintenance and Bug Fixing Work
- Continuous Delivery Environments

Extreme Programming (XP)

Overview

- Agile methodology focused on engineering practices
- Emphasizes technical excellence and frequent releases
- Core values: Communication, Simplicity, Feedback, Courage, Respect

XP Practices

- Test-Driven Development (TDD): Write tests before code
- Pair Programming: Two developers work together at one workstation
- Continuous Integration: Integrate and test code frequently
- Refactoring, Simple Design, Collective Code Ownership

XP: Pros & Cons

- Pros:
 - - High code quality and test coverage
 - - Rapid feedback and adaptability
 - - Encourages collaboration and learning
- Cons:
 - - Requires skilled developers
 - - Can be intense and exhausting
 - - Not suitable for all team cultures

Lean Software Development Principles

- Eliminate waste
- Amplify learning
- Decide as late as possible
- Deliver as fast as possible
- Empower the team
- Build integrity in
- See the whole

Toyota Production System & Lean

- Lean is inspired by Toyota Production System (TPS)
- TPS focuses on eliminating waste (muda)
- Just-In-Time production and continuous improvement (kaizen)
- Adapted to software to improve efficiency and quality

Lean Complementing Scrum & Kanban

- Lean principles enhance Agile frameworks
- Scrum: Lean helps reduce waste in Sprint planning and reviews
- Kanban: Lean reinforces flow efficiency and continuous improvement
- Combining practices leads to better agility and value delivery